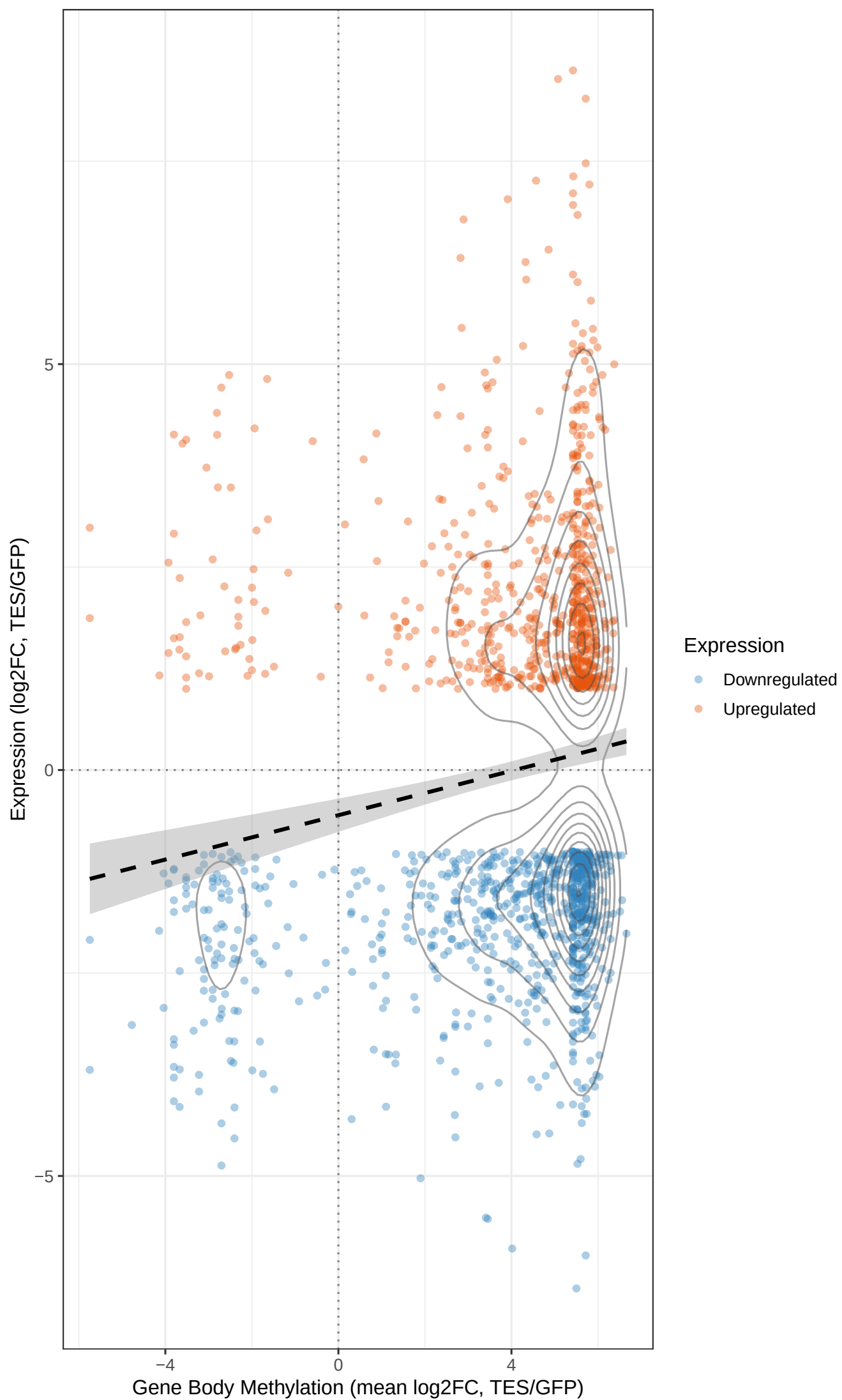


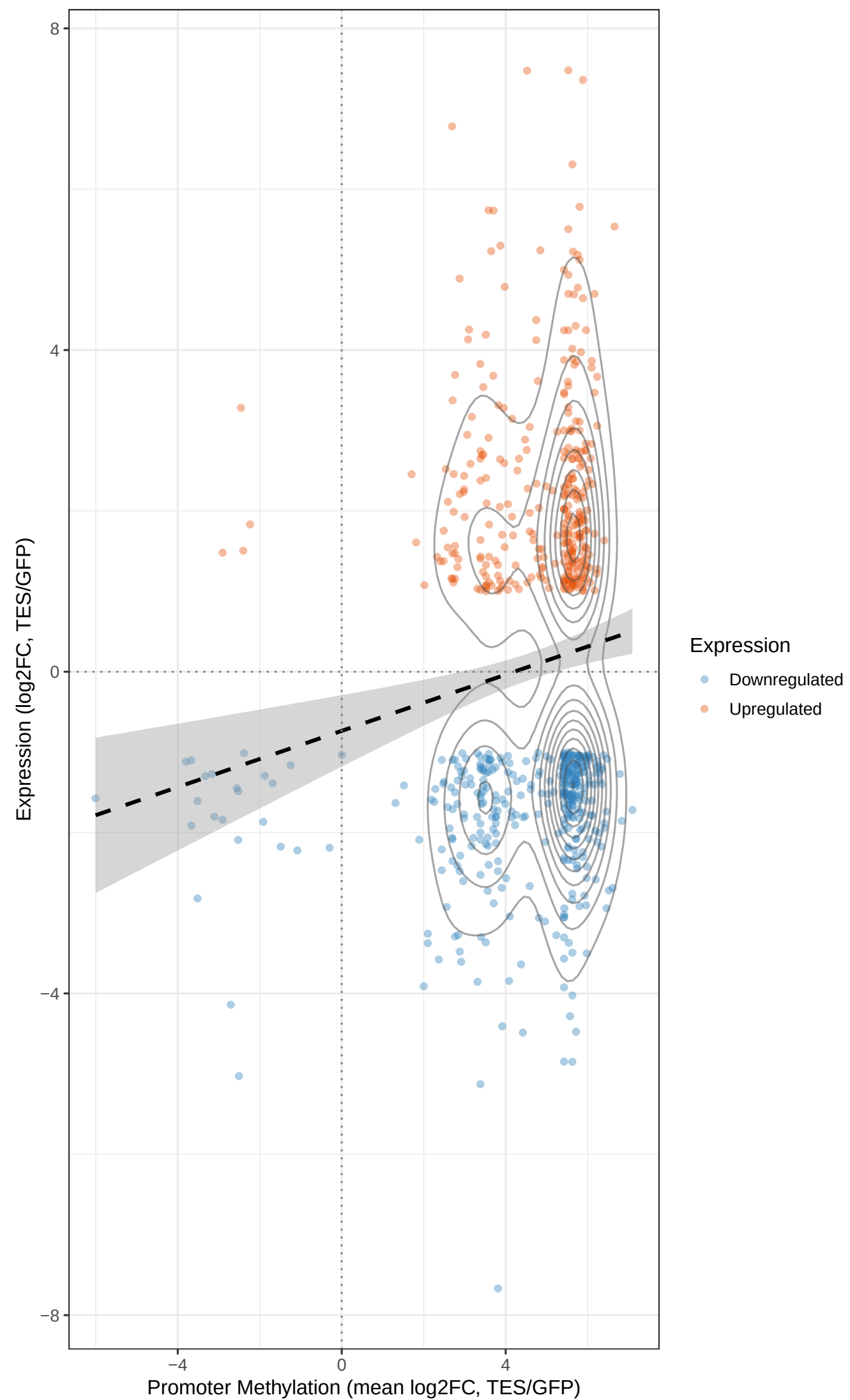
Gene Body Methylation vs Expression (DEGs)

n = 1544 | r = 0.155 | Note: Most points are in hypermethylated region (logFC > 0)



Promoter Methylation vs Expression (DEGs)

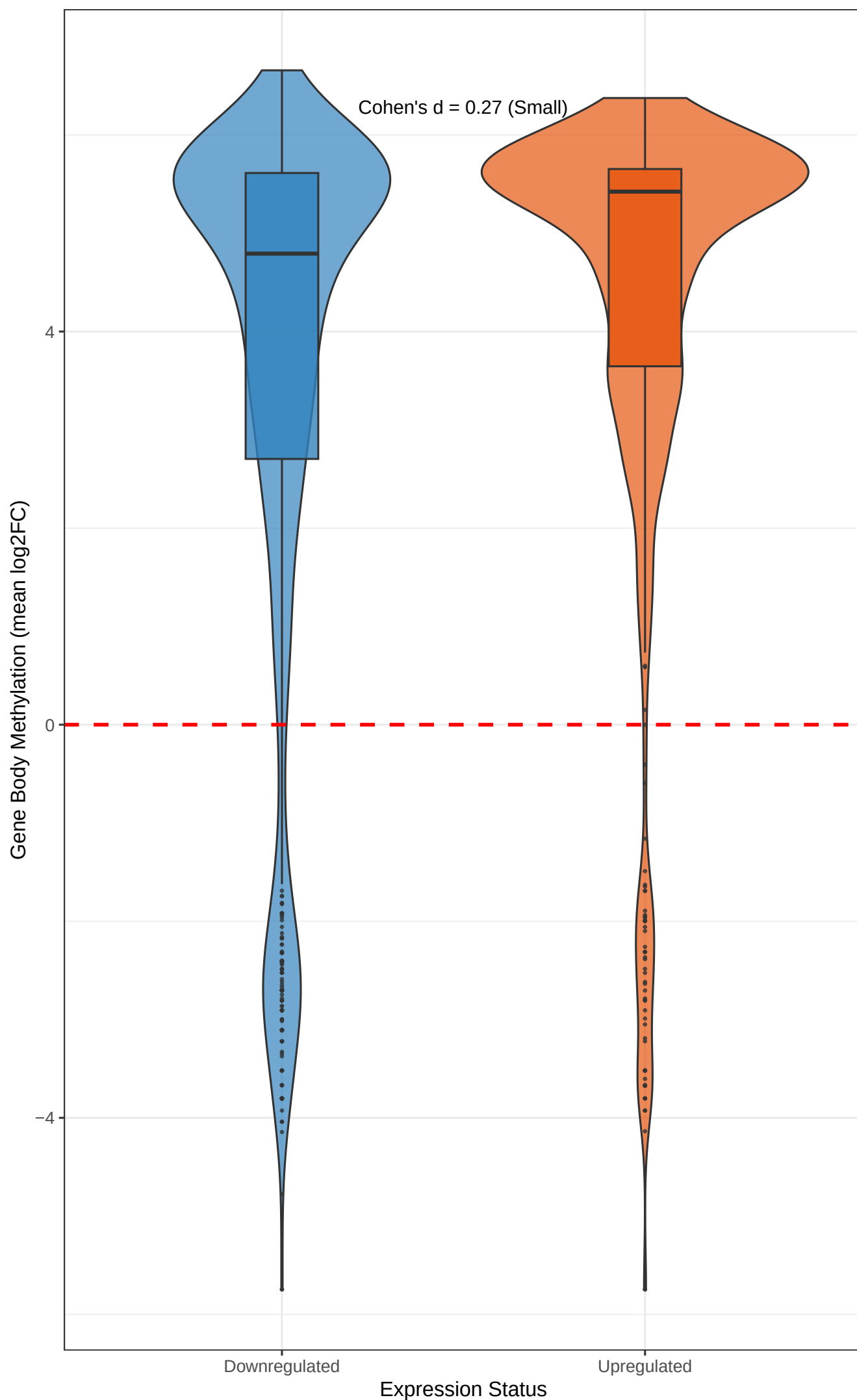
n = 710 | r = 0.142



Gene Body Methylation by Expression Status

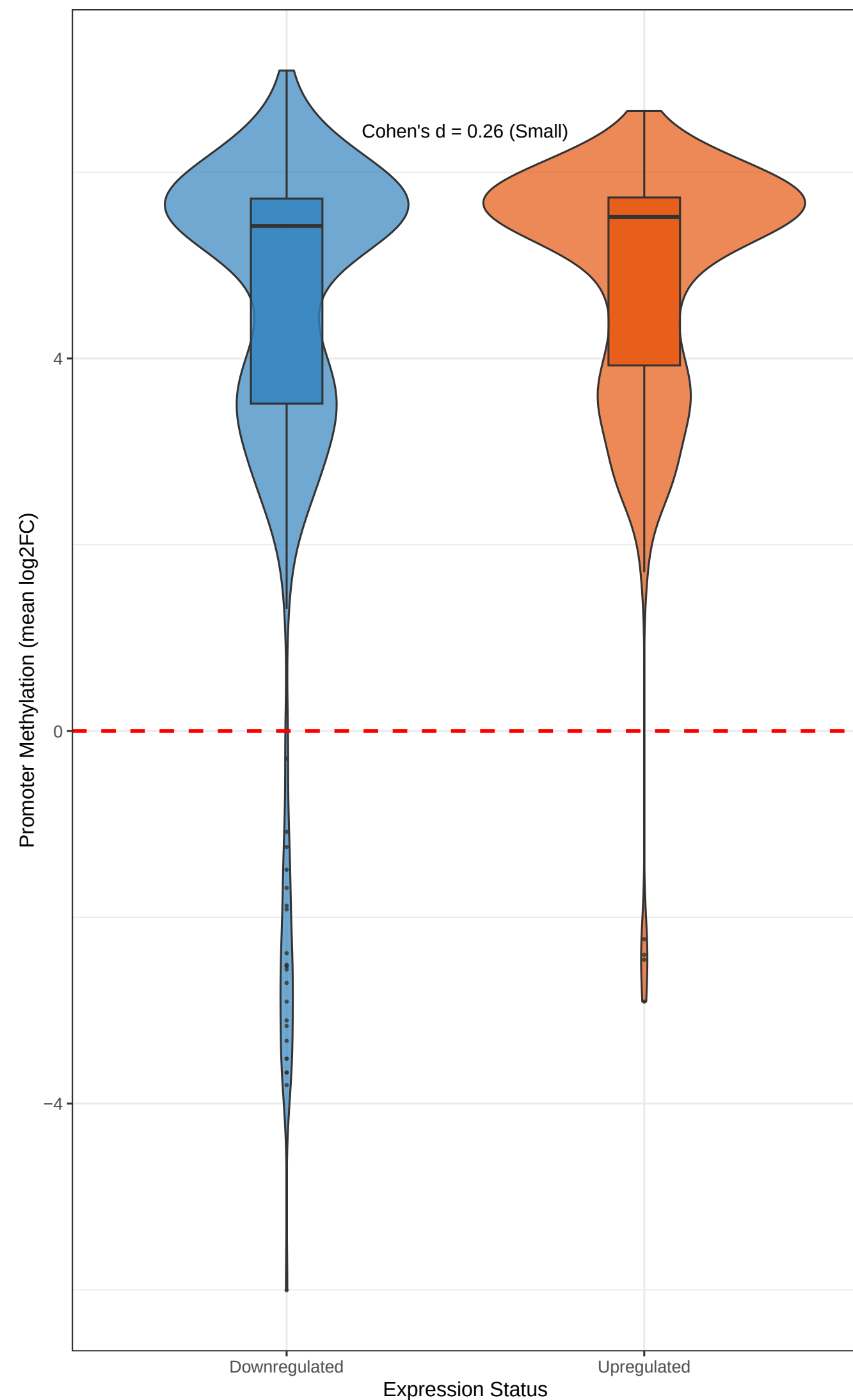
Both groups show hypermethylation (>0), but upregulated genes gain MORE

Up: mean=4.27 | Down: mean=3.56 | $p=4.91e-07$



Promoter Methylation by Expression Status

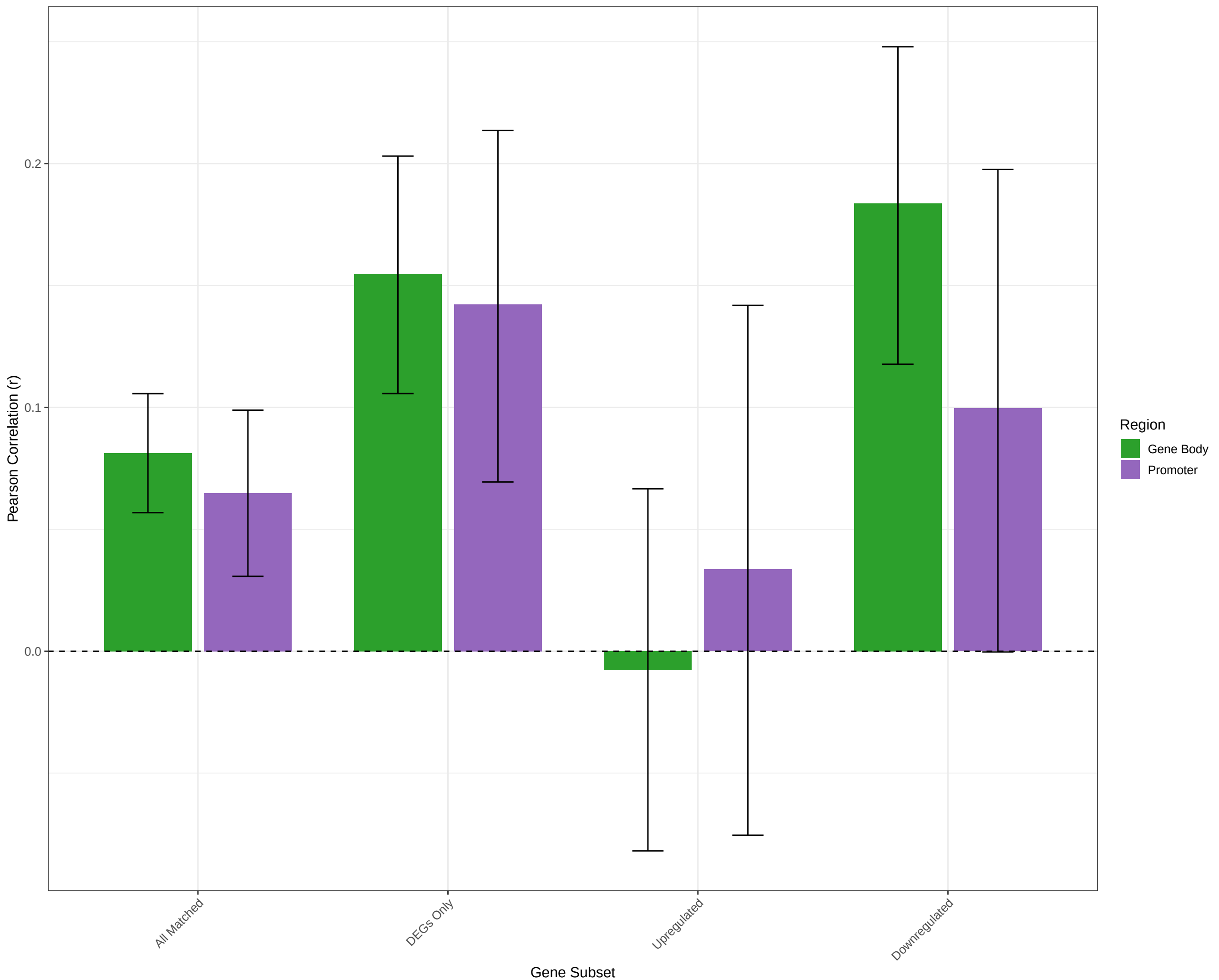
Up: mean=4.85 | Down: mean=4.37 | $p=3.59e-02$



Methylation-Expression Correlation by Region and Gene Subset

Error bars show 95% confidence intervals

Positive r = higher methylation associated with higher expression



Distribution of Gene Body Methylation in DEGs

Red dashed line = no change (logFC=0)
Note: Nearly all genes show hypermethylation (logFC > 0)

